

Python

Introduction

Python is a high-level, general-purpose programming language known for readable syntax and a large ecosystem of libraries.

Python is widely used in artificial intelligence, machine learning, data science, automation, web development, and scripting.

Python Variables and Data Types

A variable stores a value that can be used later in a program.

- int: whole numbers such as 10
- float: decimal numbers such as 3.14
- str: text such as 'Hello'
- bool: True or False
- list: an ordered collection of values
- dict: a collection of key-value pairs

Conditions and Loops

Conditional statements allow a program to make decisions using if, elif, and else.

Loops repeat code. Common Python loops are for and while.

Functions

A function is a reusable block of code that performs a specific task.

```
def greet(name):  
    return f"Hello, {name}"  
  
message = greet("Amina")  
print(message)
```

Lists and Dictionaries

Lists are useful for storing ordered collections. Dictionaries store data using keys and values.

```
languages = ["Python", "Java", "R"]  
  
student = {  
    "name": "Amina",  
    "skill": "Python"  
}
```

File Handling

Python can read and write files using open(). File handling is useful when preparing documents and datasets for AI applications.

```
with open("data.txt", "r") as file:  
    text = file.read()  
  
print(text)
```

Python in AI and RAG

Python is commonly used for RAG applications because it provides libraries for document loading, text processing, embeddings, vector databases, and language models.

A Python RAG application can load documents, split them into chunks, create embeddings, retrieve relevant information, and send the retrieved context to an LLM.